

Yichao Future Fluorescence Prediction: Current Results and Improvement Strategy

OrganoidAgent Yichao analysis

May 11, 2026

Abstract

This report documents the completed Yichao projected-organoid prediction experiments and the next technical direction. Same-time brightfield-to-fluorescence prediction ($B_t \rightarrow F_t$) is feasible: the held-out test set achieved image MAE 0.1077, expression AUROC 0.728, and fluorescence peak Pearson correlation 0.685. Future fluorescence prediction from earlier brightfield ($B_{0:k} \rightarrow F_{p>k}$) remains unsolved by the current models. The separate scalar future model achieved AUROC 0.228 and future peak Pearson 0.134. The joint model, which combines same-time B2F supervision with dense future fluorescence prediction, achieved AUROC 0.223 and future peak Pearson -0.049 on the held-out test split. Visual outputs show that predicted future fluorescence is not close to true future fluorescence. The current evidence supports same-time morphology-to-fluorescence learning, but indicates that future expression prediction needs cleaner temporal targets, stronger longitudinal supervision, and a more suitable model design.

1 Purpose

The goal was to test four progressively harder questions:

1. Can projected brightfield predict contemporaneous fluorescence?
2. Which explicit organoid morphology features explain fluorescence distinction?
3. Can early brightfield predict later fluorescence expression?
4. Can a joint model use same-time B2F as auxiliary supervision to improve future prediction?

The first two questions are supported by the current experiments. The future-prediction questions are not yet solved on held-out test data.

2 Data and Splits

The projected complete/non-edge Yichao dataset was built from:

`\outputroot/future_expression.sqlite`

The prepared dataset contains 9,380 projected complete/non-edge organoid instances, 363 linked time-series tracks, and 6,088 early-prefix future samples. The future samples are split into 4,277 train, 1,178 validation, and 633 held-out test samples.

3 Model Tasks

3.1 Same-time B2F

The same-time model learns:

$$\hat{F}_t = G_\theta(B_t), \quad (1)$$

where B_t is the projected brightfield crop and F_t is the corresponding projected fluorescence crop.

The image loss uses organoid masks:

$$\mathcal{L}_{\text{img}} = \text{mean}(|\hat{F} - F|M) + 0.2\text{mean}(|\hat{F} - F|(1 - M)). \quad (2)$$

3.2 Scalar future baseline

The scalar future model takes a brightfield prefix and morphology features:

$$P_k = \{(B_j, X_j)\}_{j=k-L+1}^k, \quad (3)$$

and predicts future-expression status, future peak fluorescence, and future fluorescence AUC:

$$(\hat{a}_p, \hat{r}_p, \hat{q}_p) = H_\phi(P_k), \quad p > k. \quad (4)$$

3.3 Joint future model

The joint model adds dense image supervision:

$$(\hat{F}_p, \hat{a}_p, \hat{r}_p, \hat{q}_p, \hat{F}_k) = J_\psi(P_k, B_k), \quad (5)$$

with objective:

$$\mathcal{L} = \mathcal{L}_{\text{img}}(\hat{F}_p, F_p, M_p) + 0.5\mathcal{L}_{\text{img}}(\hat{F}_k, F_k, M_k) + 0.25\mathcal{L}_{\text{scalar}}. \quad (6)$$

The intended intuition was reasonable: because $B_t \rightarrow F_t$ is learnable, auxiliary B2F might regularize the representation used for $B_{0:k} \rightarrow F_p$. The held-out test result shows that this auxiliary supervision alone was not enough.

4 Quantitative Results

Table 1: Final held-out test metrics. The B2F task works; both future-prediction tasks do not yet generalize.

Metric	Same-time B2F	Scalar future	Joint future
Test examples	1,088	633	633
Expression examples	14	not recorded in metric JSON	11
Image MAE	0.1077	not predicted	0.3112
Image PSNR	14.18 dB	not predicted	7.95 dB
Expression AUROC	0.728	0.228	0.223
Average precision	0.110	0.0156	0.0127
Peak Pearson	0.685	0.134	-0.049
AUC Pearson	not applicable	0.105	0.075

Interpretation. The future classifiers are worse than a useful ranking model on the held-out test set. The joint model’s dense future fluorescence prediction also does not work visually or quantitatively: image MAE is high, PSNR is low, and the prediction panels show outputs that do not match the true future fluorescence.

5 Same-Time B2F Is Feasible

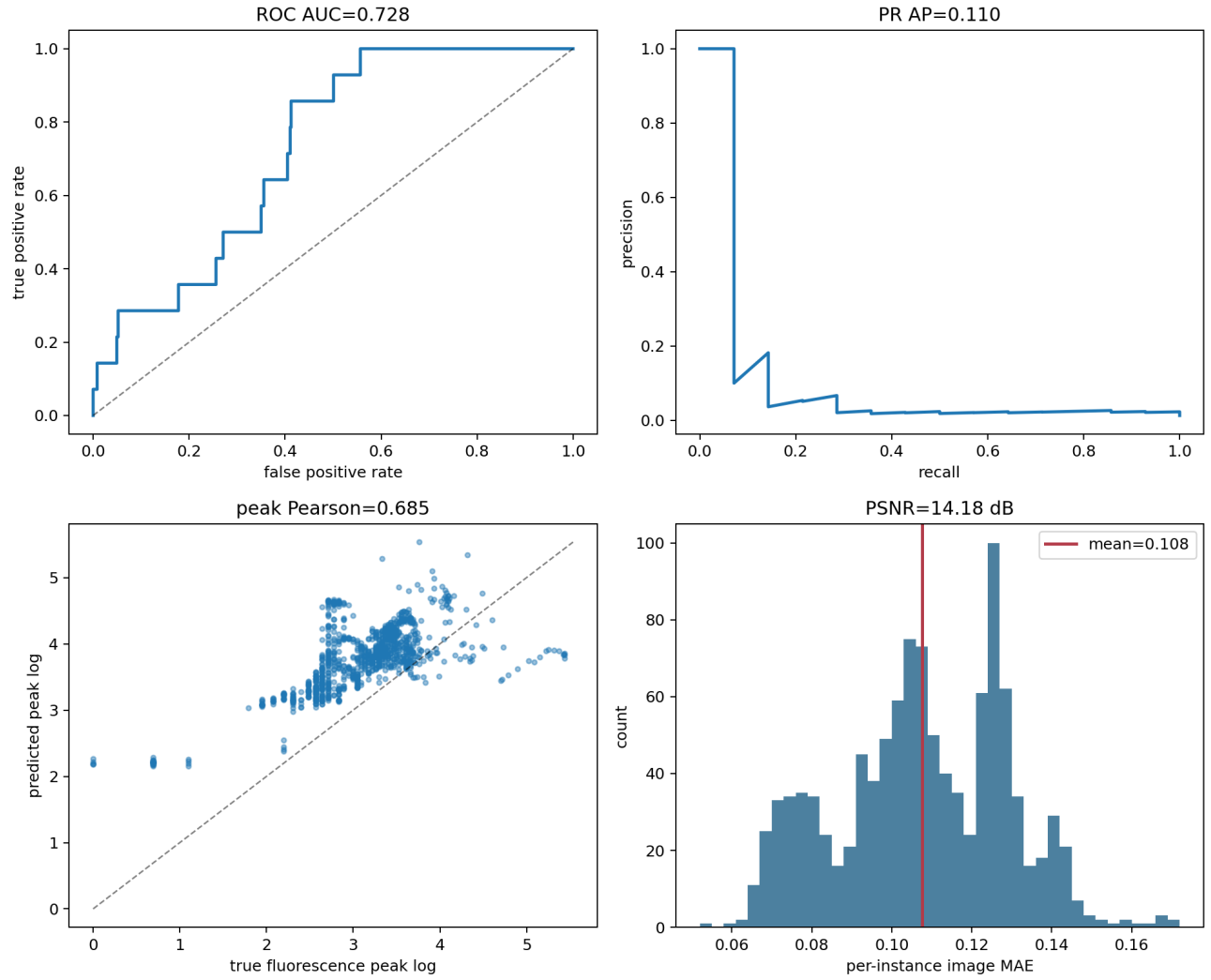


Figure 1: Same-time B2F performance summary. The predicted fluorescence intensity has meaningful held-out correlation with true fluorescence.

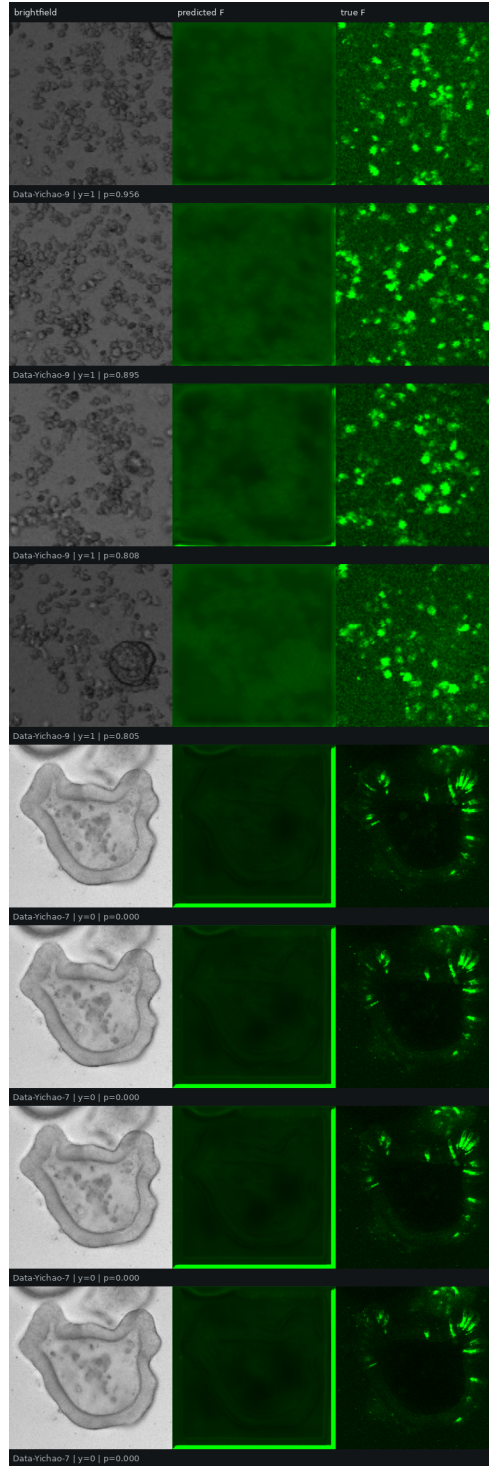


Figure 2: Held-out same-time B2F prediction examples. Columns show brightfield input, predicted fluorescence, and true fluorescence.

The B2F result establishes that brightfield morphology contains contemporaneous fluorescence-related information. This does not imply that early brightfield contains enough information to predict future expression.

6 Feature Explanation

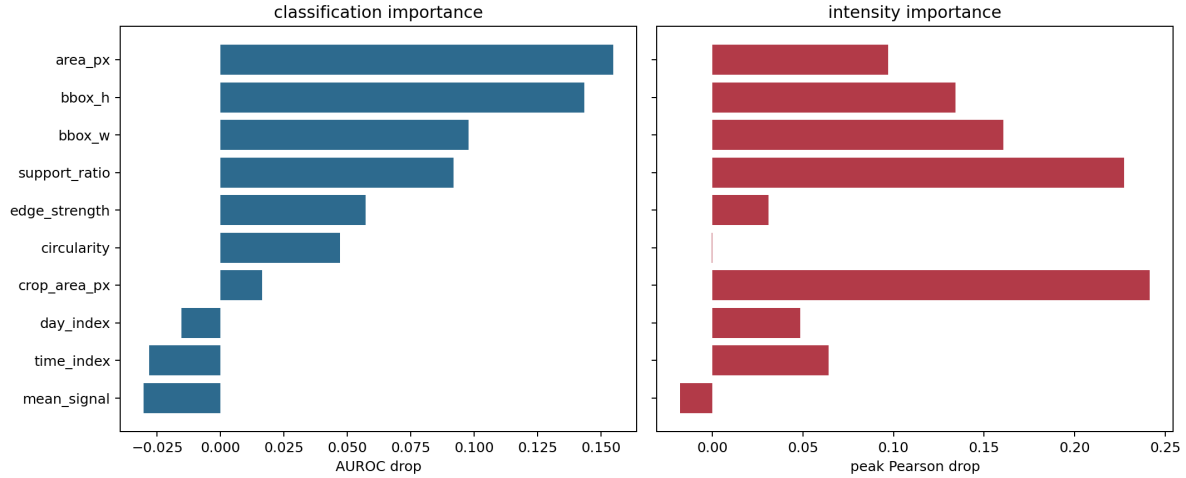


Figure 3: Explicit feature explanation for fluorescence distinction. Size and morphology features are the strongest signals.

Table 2: Top explanatory features from same-time feature analysis.

Feature	AUROC drop	Peak-correlation drop
Area	0.155	0.097
Bounding-box height	0.143	0.134
Bounding-box width	0.098	0.161
Support ratio	0.092	0.227
Edge strength	0.057	0.031
Circularity	0.047	-0.000

These features are plausible and useful for same-time fluorescence distinction. They mainly describe organoid size, projected shape, segmentation support, and boundary quality. They are not sufficient evidence that future differentiation can be predicted before expression.

7 Scalar Future Prediction Fails

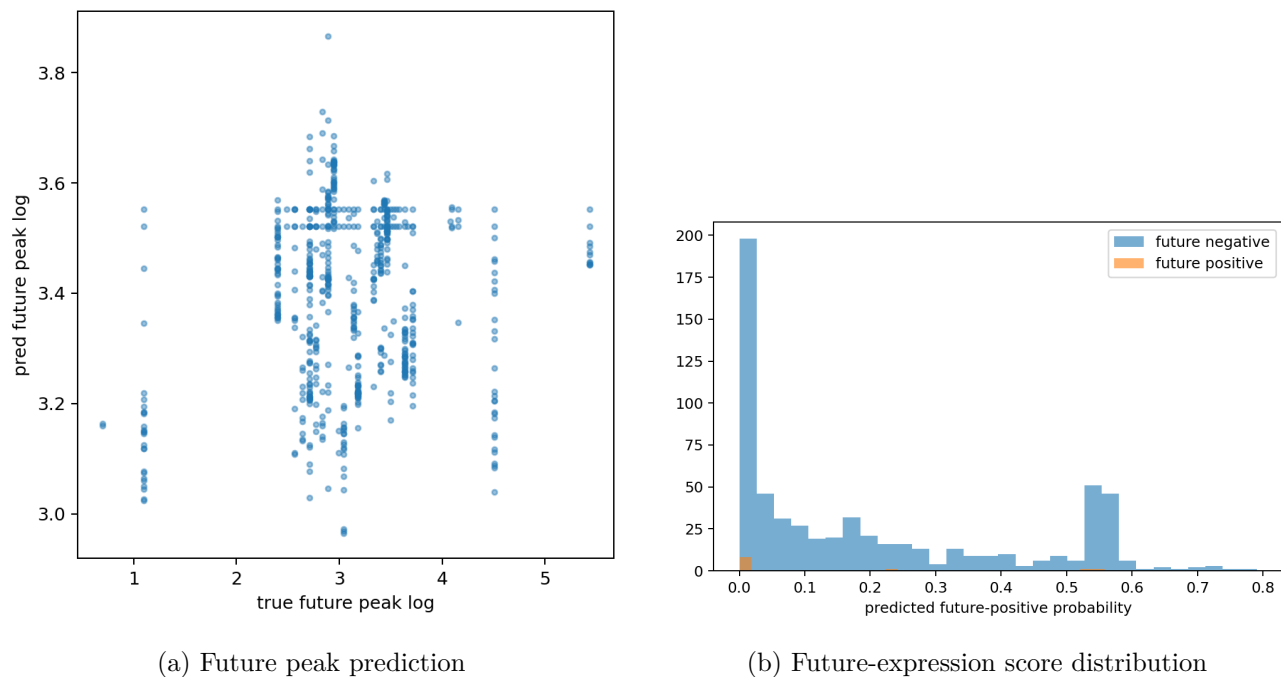


Figure 4: Scalar future baseline outputs. The prediction is not close enough to truth for useful future expression prediction.

The scalar future model is a weak baseline. It does not separate future-expressing from non-expressing examples and it does not predict the future fluorescence peak accurately enough to support biological inference.

8 Joint Future Prediction

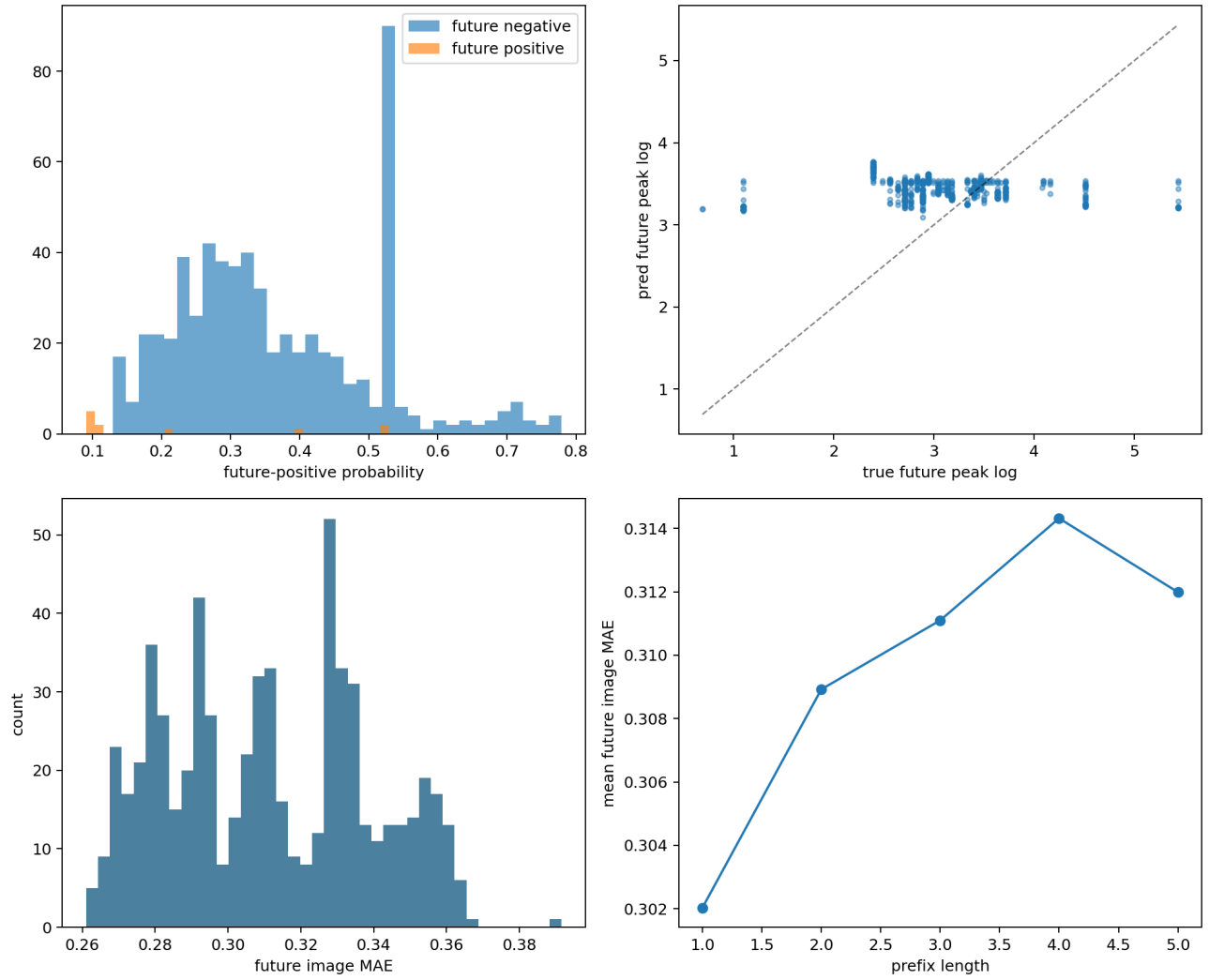


Figure 5: Joint model final test summary. Validation appeared promising during training, but the selected model does not generalize on the held-out future test split.

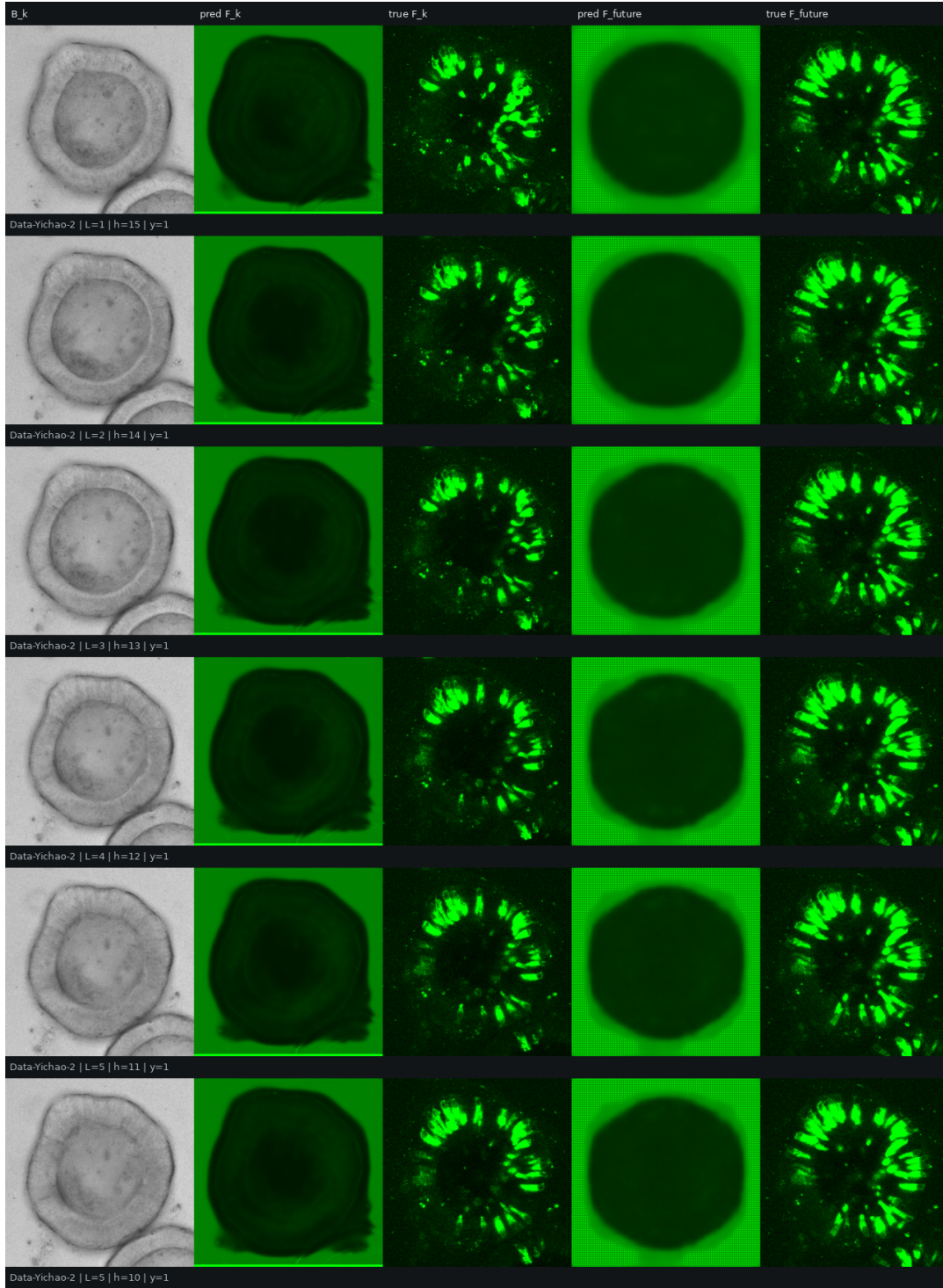


Figure 6: Joint model held-out test prediction panel. Columns show current brightfield, predicted current fluorescence, true current fluorescence, predicted future fluorescence, and true future fluorescence. This panel is the most direct visual evidence that the future prediction is not close to the truth.

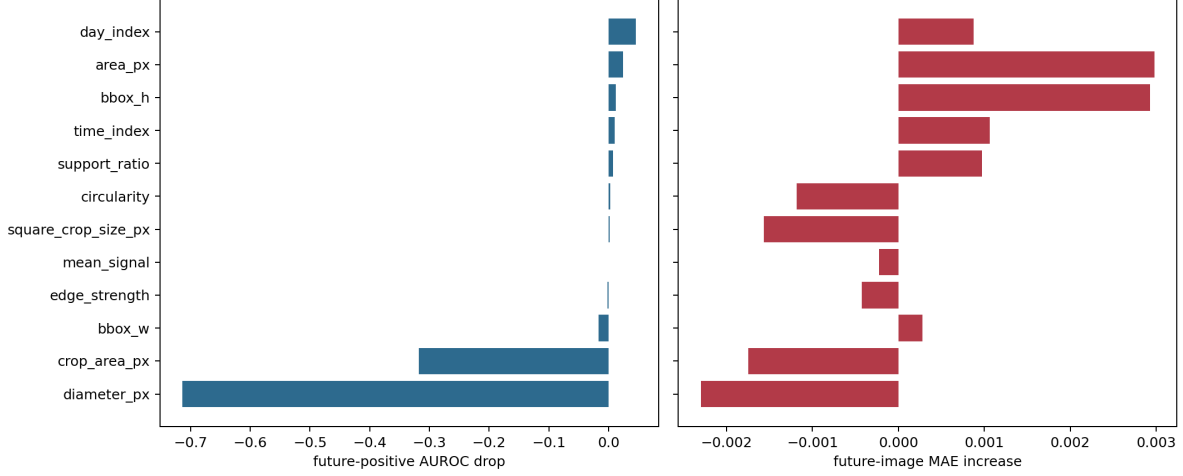


Figure 7: Joint model feature ablation. Some temporal/size features affect the weak classifier, but the overall held-out future prediction remains poor.

The joint model did not solve the future task. The most important observation is not just the weak scalar metrics, but the visual mismatch in Fig. 6: the predicted future fluorescence is not close to the true future fluorescence.

9 Likely Failure Causes

The current result is technically useful. It narrows the problem:

1. **Same-time signal exists, but future signal may not be present early enough.** B2F learns $B_t \rightarrow F_t$, but this does not guarantee $B_{0:k} \rightarrow F_p$.
2. **Future-expression examples are rare.** The held-out joint test split has only 11 future-expression examples.
3. **Track linking is approximate.** Any wrong organoid association across time directly corrupts future targets.
4. **Last-frame target may be a poor biological endpoint.** The last available future frame may not equal the biologically most informative or peak-expression frame.
5. **Dataset/domain split effects are strong.** Validation performance looked better than held-out test performance, indicating nontrivial distribution shift.

10 Model Improvement Strategy

The current result does not justify only scaling the current network. A larger model can overfit the same weak temporal supervision more easily. The better direction is a staged model with stronger temporal targets:

1. **Keep B2F as pretraining and auxiliary supervision.** Train an image model on $B_t \rightarrow F_t$ first because that task has demonstrated signal. Use the encoder weights for future prediction instead of learning future representations from scratch.

2. **Use joint fine-tuning, not pure separate training.** Separate B2F is useful for representation learning, but the final future model should be joint: current fluorescence reconstruction regularizes the encoder while the future head learns $B_{0:k} \rightarrow F_p$.
3. **Increase model capacity only after target cleanup.** A U-Net/ResNet hybrid or ConvNeXt-style encoder with temporal pooling is reasonable, but the first bottleneck is likely target noise and rare expression examples, not insufficient parameter count.
4. **Use larger image sizes when feasible.** Default 256×256 is useful for speed and baseline comparison, but elongated peripheral fluorescent cells and internal void morphology may be compressed too much. The next run should test 384×384 or 512×512 crops with mixed precision and gradient accumulation.
5. **Predict biologically better targets.** Replace only-last-frame prediction with peak future expression, first-expression time, and horizon-specific targets such as $D_k \rightarrow D_{k+1}$, $D_k \rightarrow D_{k+2}$, and $D_k \rightarrow D_{\text{peak}}$.
6. **Balance expression examples.** Oversample rare future-expression tracks and use ranking/contrastive losses so the model learns which early morphologies precede later expression rather than being dominated by non-expressing examples.

The most practical next architecture is:

$$z_k = \text{TemporalEncoder}(\text{ImageEncoder}(B_1), \dots, \text{ImageEncoder}(B_k), X_{1:k}), \quad (7)$$

$$(\hat{F}_k, \hat{F}_p, \hat{y}_p, \hat{s}_p) = \text{DecoderHeads}(z_k), \quad (8)$$

where $\hat{F}_k$ is current fluorescence reconstruction, $\hat{F}_p$ is future fluorescence, $\hat{y}_p$ is future-expression probability, and $\hat{s}_p$ is scalar future intensity. This keeps the useful B2F signal while forcing the shared representation to serve the future task.

11 Recommended Data Step

Do not increase model complexity before checking the temporal labels. The next scientifically realistic step is to improve the data target:

1. Manually verify a subset of linked tracks, especially fluorescence-expressing trajectories.
2. Train/evaluate by horizon: short future, medium future, long future.
3. Compare `last_future` targets with `peak_future` targets.
4. Build a cleaner expression-control dataset where Yichao confirms which structures are true fluorescent cells versus debris/background.
5. Keep same-time B2F as a validated auxiliary or pretraining task, but do not claim future prediction until horizon-specific held-out tests improve.

12 Reproducibility

Main completed output folders:

- Same-time B2F: `\outputroot/stage1_b2f_evaluation`
- Scalar future baseline: `\outputroot/stage2_future_expression`
- Joint future model: `\outputroot/stage3_joint_b2f_future`
- Previous consolidated report: `\outputroot/previous_run_evaluation`

Main code folder:

`/home/lachlan/ProjectsLFS/OrganoidAgent/differentiation_prediction/yichao_future_expression`

13 Conclusion

The current conclusion is:

The current Yichao projected dataset supports same-time brightfield-to-fluorescence prediction and morphology-based explanation, but it does not yet support reliable future fluorescence expression prediction from earlier brightfield. The report includes both separate training results and joint training results; the joint model is the most relevant current future-prediction result, and it still does not match true future fluorescence on held-out examples.

This is still useful because it prevents overclaiming and clarifies the next experimental requirement: cleaner longitudinal tracking, expression-control examples, horizon-specific targets, and a stronger temporal model initialized from the successful B2F task.

References

- [1] Isola, P., Zhu, J.-Y., Zhou, T. and Efros, A. A. Image-to-image translation with conditional adversarial networks. *CVPR* (2017). <https://arxiv.org/abs/1611.07004>
- [2] Allen Cell Modeling. Pytorch-FNet: label-free prediction of fluorescence images from transmitted-light microscopy. https://allencellmodeling.github.io/pytorch_fnet/
- [3] Vafaeikia, P., Namdar, K. and Khalvati, F. A brief review of deep multi-task learning and auxiliary task learning. <https://arxiv.org/abs/2007.01126>